TASK 08 – PLAN THE ALGORITHM BEFORE CODING

Aim

To plan the milk payment calculator using algorithmic problem-solving steps, pseudocode, a flowchart and a hand trace.

1. Steps of Algorithmic Problem Solving

The four main steps are:

1. Understand the problem

Understand the inputs required: litres of milk, fat percentage and SNF percentage. The output is the payment amount for each farmer and the total payment for ten days.

2. Plan the solution

Prepare the algorithm, pseudocode and flowchart before writing the Python program.

3. Write the program

Convert the planned algorithm into Python functions such as rate_per_litre() and payout().

4. Test the program

Test normal entries, values above and below the standard rate, boundary values and invalid entries. Compare the actual results with the expected results.

2. Pseudocode

ALGORITHM 1: Calculate One Farmer's Payment

START

INPUT litres

INPUT fat

INPUT snf

IF litres is valid AND fat is within the allowed range AND snf is valid THEN

SET fat_steps TO round((fat - 3.5) / 0.1)

SET snf_steps TO round((snf - 8.5) / 0.1)

SET rate TO 30.00 + fat_steps * 0.40 + snf_steps * 0.30

SET rate TO round(rate, 2)

SET payment TO round(litres * rate, 2)

OUTPUT rate

OUTPUT payment

ELSE

OUTPUT "Invalid entry"

ENDIF

END

ALGORITHM 2: Calculate Ten-Day Total

START

SET total TO 0

INPUT number_of_entries

FOR each entry FROM 1 TO number_of_entries

INPUT litres, fat, snf

IF entry is valid THEN

SET payment TO payout(litres, fat, snf)

SET total TO total + payment

ELSE

OUTPUT "Invalid entry"

ENDIF

ENDFOR

SET total TO round(total, 2)

OUTPUT total

END

3. Flowchart for Validating One Entry

The flowchart checks whether the quantity, fat and SNF values are valid before calculating payment.

1. Start.
2. Read litres, fat and SNF.
3. Check whether the values are valid.
4. If valid, calculate the rate and payment.
5. Display the payment.
6. If invalid, display an error message.
7. End.

4. Hand Trace for Three Farmer Entries

Given rate rules:

- Base rate = Rs. 30.00
- Standard fat = 3.5%
- Standard SNF = 8.5%
- Every 0.1% fat step = Rs. 0.40
- Every 0.1% SNF step = Rs. 0.30

Farmer 1

Input:

- Litres = 10
- Fat = 3.5%
- SNF = 8.5%

Fat steps = 0

SNF steps = 0

Rate = 30.00 + (0 × 0.40) + (0 × 0.30)

Rate = Rs. 30.00

Payment = 10 × 30.00 = **Rs. 300.00**

Farmer 2

Input:

- Litres = 8.5

- Fat = 4.2%
- SNF = 8.7%

Fat steps = 7

SNF steps = 2

Rate = 30.00 + (7 × 0.40) + (2 × 0.30)

Rate = Rs. 33.40

Payment = 8.5 × 33.40 = **Rs. 283.90**

Farmer 3

Input:

- Litre